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Occasional Paper

Space Domain Awareness as a Foundational National Security Issue

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November 5, 2024

- The exponential growth of space capabilities across the commercial, national security, and civil sectors is driving economic, military, intelligence, and scientific advances.
- With history as the teacher, the new race for control of this domain, through competition with China, Russia, and other countries, inevitably will lead to eventual conflict as countries race for control of strategic locations and exploitation of the associated resources.
- Space domain awareness (SDA) leads to knowledge of activities throughout the space domain (orbital, cislunar, lunar, and Martian). Today the U.S. not only lacks full space domain awareness but is lagging behind China.
- The U.S. Government needs to lead the world in monitoring, collecting, analyzing and understanding the space domain in near-real time to provide both decision advantage and enabling information for all sectors of space.

In the last two decades the number of space-faring countries has nearly doubled, with 16 of those countries possessing independent space launch capability. Likewise, the number of commercial space entities has risen from just a handful in the early 2000's to over 10,000 companies globally with a market value of space-focused companies over $4T (in 2021).[1] And the number of active on-orbit satellites has increased by nearly a factor of 20 to almost 10,000.[2] The annual global space economy is estimated to rise from $400B in 2023 to nearly $1.8T in 2035[3], while the ability to conduct global diplomatic, intelligence, military and economic activities would be crippled without an effective SDA architecture.

Economic advantage alone is sufficient reason for a space race. With advances in robotics, remote sensing, and autonomous operations, everything from scientific discoveries to information dominance to resource mining is driving the commercial industry. All aspects of modern life are enabled through space capabilities. The country that dominates the space domain will drive international standards, norms, claim resources, and control the market. China's leaders have made clear this is their strategic intent.

Understanding the environment, including the activities of strategic competitors and the consequences of their actions, let alone any military actions, requires robust and resilient space domain awareness. SDA turns collected data across all phenomenologies into actionable intelligence and information. The resulting operational intelligence needs to be timely, actionable and shareable across the national security, commercial, civil and scientific sectors, to include allies and partners.

1 John Koetsier, "Space Inc: 10,000 Companies, $4T Value, and 52% American", Forbes, May 22, 2021, https://www.forbes.com/sites/johnkoetsier/2021/05/22/space-inc-10000-companies-4t-value--and-52-american/#:~:text=The%20U.S.%20now%20has%205%2C582,more%20than%2010%2C000%20total%2C%20globally.

2 "State of the Satellite Industry Report 2024", Satellite Industry Association, https://sia.org/commercial-satellite-industry-continues-historic-growth-dominating-global-space-business-27th-annual-state-of-the-satellite-industry-report/

3 Ibid.

Currently, the U.S. Government's ability to persistently and accurately see into higher orbits and beyond (including to the Moon and Mars) and turn that scant data into actionable intelligence lags China's as evidenced by Beijing's geostationary (GEO) operations and recent open source reporting[4] of China's plans to build out a lunar position, navigation and timing constellation setting them up to establish the international standard. This lack of persistent SDA coverage in higher orbits is a major shortfall and sets us up for a strategic surprise.

Low earth orbit (LEO) is also a concern from a safety-of-flight perspective. There are no compelling international norms or legal frameworks that govern traffic management in this increasingly crowded orbit. Furthermore, it is unlikely that America's primary adversaries will comply with any future norms or legal frameworks that emerge from international negotiations. Hence, robust SDA is needed not only to warn of possible conjunctions but also to detect and respond to threatening on-orbit behaviors by America's adversaries.

The U.S. Government needs to get serious about SDA. Specifically, it needs to:

1. Conduct a whole-of-government and commercial capabilities and gaps assessment of SDA.
2. Develop an integrated capabilities, data exploitation, and information sharing roadmap for SDA covering LEO through GEO orbits, lunar orbits, and Martian orbits commensurate with current plans for space exploration and development.
3. Develop an integrated budget submission to enact the roadmap.
4. Name a single executive agent to execute.
5. Redouble efforts to generate space traffic management protocols, initially in low earth orbit but eventually expanding to higher earth orbits and beyond. Such protocols can help promote safe and appropriate norms of behavior among like-minded nations and companies; however, it is unlikely China and Russia will adopt or comply with such Western-conceived standards and protocols.

Dr. Sean Kirkpatrick and Gen. Willie Shelton serve as distinguished members of the NSSA Board of Advisors.

4 See, for example, Ling Xin, "A BeiDou-like satnav system for the moon? Chinese scientists plot a possible route", South China Morning Post, July 14, 2024, https://www.scmp.com/news/china/science/article/3270195/satnav-system-moon-chinese-scientists-plot-possible-route and Andrew Jones, "China to launch lunar navigation and communications test satellites", SpaceNews, February 7, 2024, https://spacenews.com/china-to-launch-lunar-navigation-and-communications-test-satellites/